CMSC 420: Skip Lists

Justin Wyss-Gallifent

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1 Problems **2**

1 Problems

1. Suppose the keys 30,10,80,50,20 are inserted into a randomized skip list in that order.

   We know each element exists in level 0. When determine whether each element extends to the next higher level we use a random number generator which outputs either 0, which means the element extends no further, or 1, which means the element extends another level and we must check again.

   Draw the resulting randomized skip list under the assumption that the random number generator outputs the following:

   1,1,1,1,0,1,0,1,1,0,0,1,0

2. Suppose a skip list has 100 nodes. What is the probability that at least one node will reach a level of 5 or beyond?

3. This problem digs deeper into the probabilities associated with a skip list with n nodes reaching a certain number of levels.

   (a) For each $k = 1, 2, 3, ...$ what is the probability that a skip list does not reach level k? Justify.

   (b) For each $k = 0, 1, 2, 3, 4, ...$ what is the probability that a skip list does reach level greater than or equal to k? Justify.

   (c) For each $k = 0, 1, 2, 3, 4, ...$ what is the probability that the skip list reaches exactly level k? Justify.

4. When building a skip list recall that there is a 0.5 probability that a node which reaches level i also reaches level $i + 1$. Suppose we change this probability to some $0 < p < 1$.

   If $n = 80$ nodes are inserted into such a skip list what is the maximum value of p which ensures that the probability that at least one node reaches level 10 or beyond is 0.4 or below?

5. Suppose we build a skip list but we modify the probability such that each node has a p probability of extending to the next level.

   If the skip list has exactly one node, what do we expect the maximum level to be? Justify.

   Note: It may be useful to know that:

$$\sum_{i=0}^{\infty} ix^i = \frac{x}{(1-x)^2}$$

6. For a skip list containing n nodes and having enforced maximum level M, what is the probability that every node in the skip list reaches level M?

7. Suppose a skip list has n nodes and $p = 0.5$. What is the probability that the skip list reaches each of the following.

(a) Level 0 and Perhaps Further	
(b) Level $k = 0, 1, 2, ...$ and Perhaps Further	
(c) Exactly Level $k = 0, 1, 2, ...$	

8. Considering a single node in a skip list with probability $0 < p < 1$. For all $i \geq 0$, what is the probability it reaches level i but not futher?

9. Suppose the keys 5,21,17,7,10,42 are inserted into a randomized skip list in that order.

   We know each element exists in level 0. When determine whether each element extends to the next higher level we use a random number generator which outputs either 0, which means the element extends no further, or 1, which means the element extends another level and we must check again.

   Draw the resulting randomized skip list under the assumption that the random number generator outputs the following:

   1,1,0,1,0,0,1,1,1,1,0,1,1,1,0,1,1,0,

10. We proved in class that if L is the maximum level in a skip list with n nodes then $E(L) = \mathcal{O}(\lg n)$. We proved the case where $p = 0.5$. Modify this for an unknown $0 < p < 1$. You may assume the definitions of I_i and n_i and you may assume that:

$$E(n_i) = np^i$$

Start at the point:

$$E(L) = \sum_{i=0}^{\infty} E(I_i)$$

11. It can be shown that the expected maximum level in a skip list with n nodes and with $p = 0.5$ actually satisfies:

$$E(L) = \sum_{k=0}^{\infty} k \left[\left(1 - 0.5^{k+1}\right)^n - \left(1 - 0.5^k\right)^n \right]$$

This is very difficult sum to evaluate, so we won't. But however you like, evaluate or approximate this value for $n = 1000, 2000, 3000, ..., 10000$ to the nearest single digit after the decimal point.

12. Suppose a skip list with unknown $0 < p < 1$ has maximum enforced level $M \geq 0$. Let S_i be the number of steps going left in level i when following skip list search in reverse. We know that $E(S_i) \leq np^i$ for all $0 \leq i \leq M$. Use this single fact to provide a quick proof that:

$$E(S) = \mathcal{O}(n)$$

13. Suppose a skip list with $p = 0.5$ has maximum enforced level $M = \alpha \lg n$ where $\alpha \geq 0$. Let S_i be the number of steps going left in level i when following skip list search in reverse. Prove that the total number of steps S_L going left satisfies:

$$S_L = \mathcal{O}(\lg n + n^{1-\alpha})$$

14. We proved that if S_i is the number of steps going left in level i when following skip list search in reverse satisfies $E(S_i) \leq 1$ when $p = 0.5$. This is for all i when no maximum enforced level is set. Construct a version of this for an unknown p. You may need to look up an unfamiliar series for this.

15. Consider the following skip list:

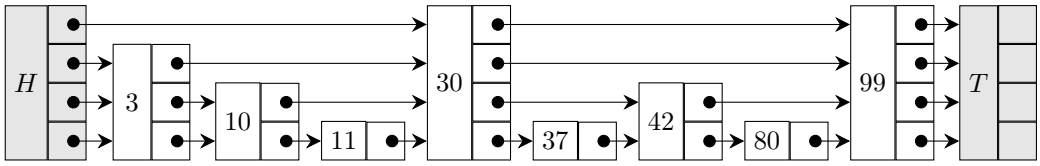

Figure 1: Skip List for the Problem

We know that search starts at the top of the header node and a *step* is either going down one pointer level or following a pointer. So for example it takes 5 steps to find 37 since we go F, D, D, D, F where F means follow a pointer and D means go down one pointer level.

(a) How many steps does it take to find 42?

How Many?	

(b) If a key is chosen uniformly at random what is the expected number of steps it will take to find it? Show your work and simplify.

16. Suppose a skip list has $p = 0.5$ and n nodes. No maximum enforced level.

(a) Suppose $n = 10$. What is the probability that at least one node reaches level 3 or higher? You do not need to simplify.

Probability	

(b) What is the minimum number of nodes that must be inserted so that the probability that at least one node reaches level 2 or higher is at least 0.7? No maximum enforced level. Simplify as much as possible; you can leave a logarithm in your answer.

Minimum Number	

17. Suppose a skip list has $p = 0.5$ and n nodes. No maximum enforced level. Calculate the probability in terms of n (unknown) that every node has exactly the same height. Show your work and simplify your answer to get rid of any Σ sums.